

Amazon's Fine Food Product Reviews

DataSci 207 Final Project

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Motivation for NLP Sentiment Analysis



Textual information, such as reviews, are widely available

Sentiment analysis can help automate this process

Tedious and repetitive to have to look through manually

Evaluation of reliability of reviews

Can be used to quickly identify public opinion on items, and may be combined with summarization techniques to quickly identify key points across many reviews

Data: Amazon's Fine Food Product Reviews

Overview

Total Reviews: Over 500,000

Total Users: Around 256,000

Total Products: Approximately 74,000

Purpose: Analyze consumer opinions, develop recommendation systems, study review patterns

[Link to Data](#)

Features

User ID: Unique identifier for each user

Product ID: Unique identifier for each product

Profile Name: User's profile name

Helpfulness Rating: Helpfulness of the review as voted by users

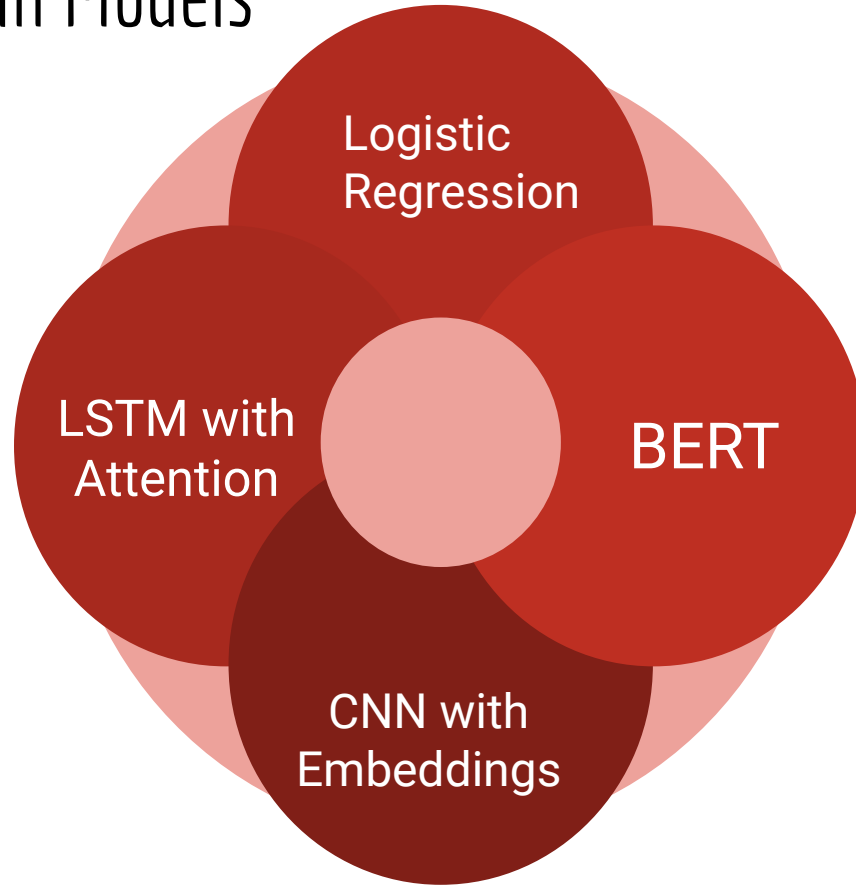
Score: Rating given by the user (1 to 5 stars)

Time: Timestamp of the review

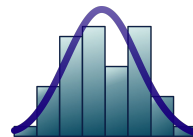
Summary: Short summary of the review

Text: Full text of the review

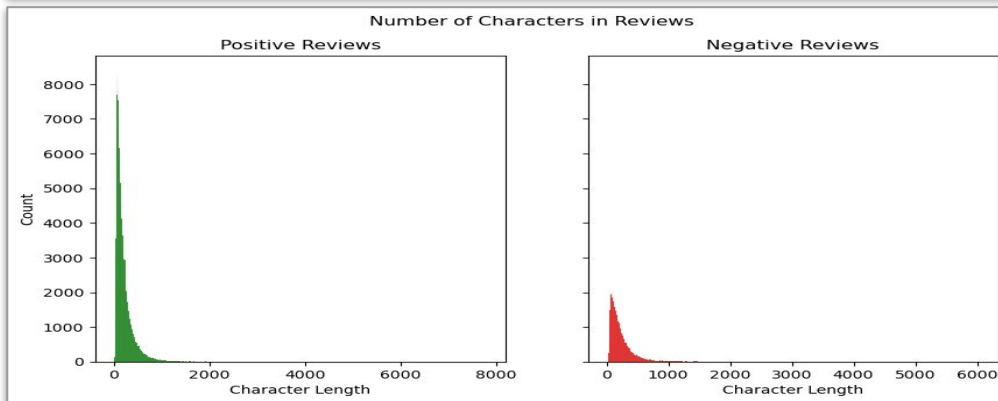
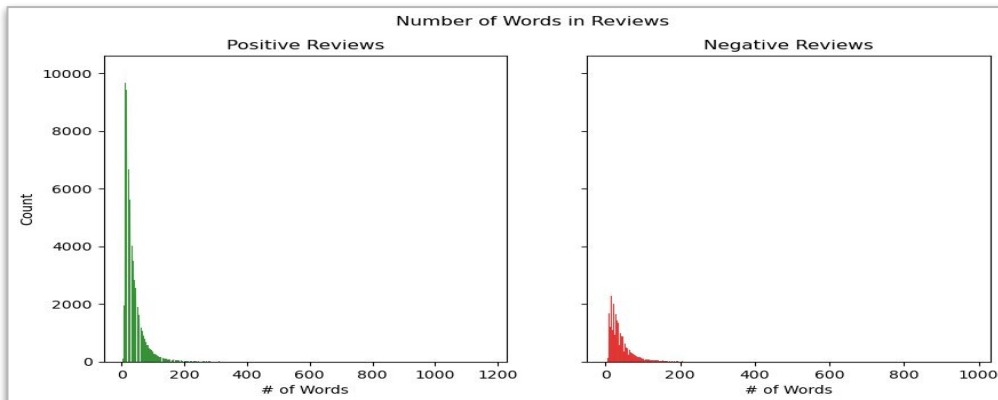
Workflow - 4 Main Models



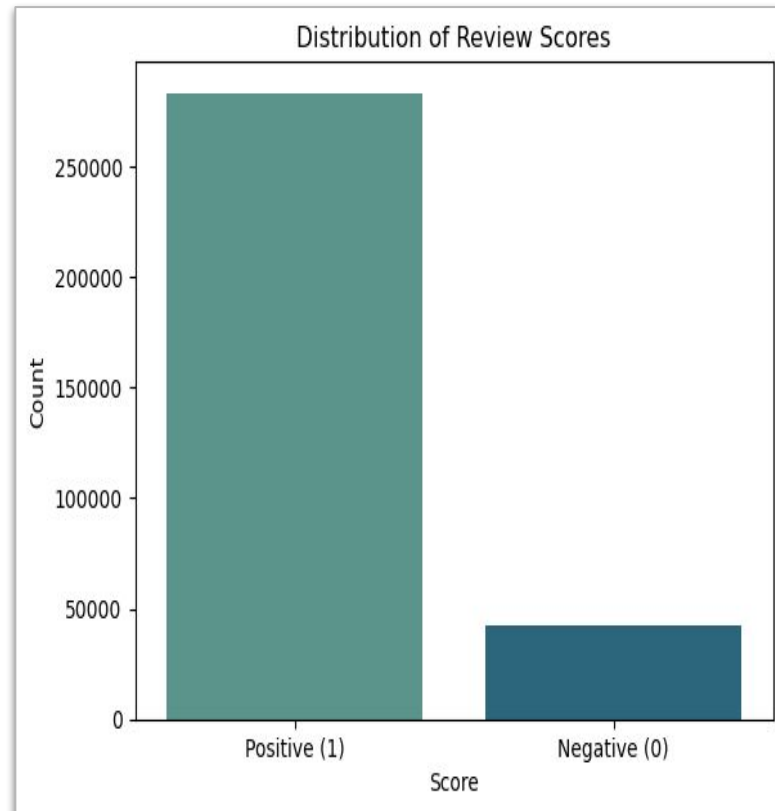
Data Distribution



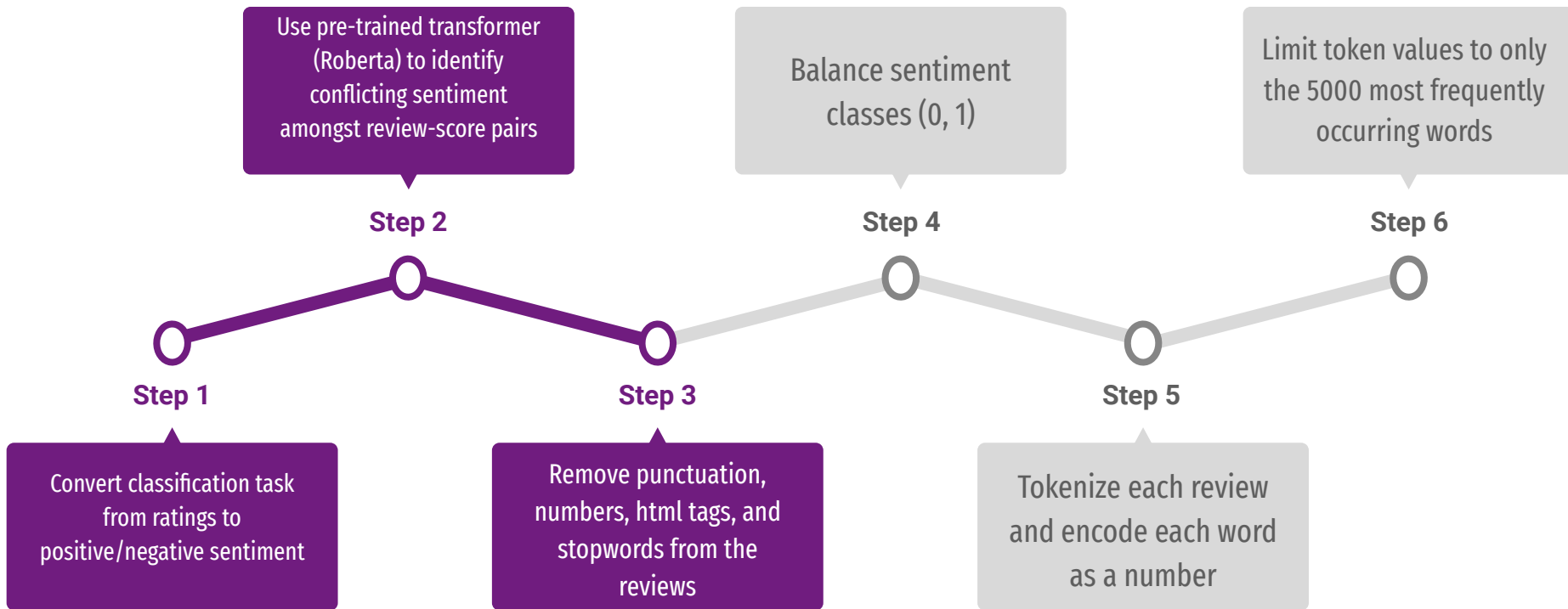
Character and Word Length



Review Score

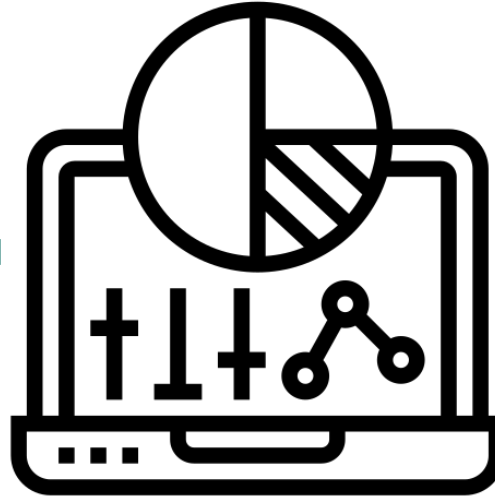


Data Preprocessing



Baseline Model

Used a base
decision tree as
our baseline
since it can
handle
imbalances well



Ran it on base,
unbalanced
data

Baseline Model Classification Report



Validation Accuracy: 0.7830570902394107

Validation Classification Report:

	precision	recall	f1-score	support
0	0.56	0.01	0.02	17118
1	0.78	1.00	0.88	61617
accuracy			0.78	78735
macro avg	0.67	0.50	0.45	78735
weighted avg	0.73	0.78	0.69	78735

Logistic Regression - Model 1



Step 1: N-Grams Implementation

TF-Vectorizer with n-grams

- Understand the context in which words appear
- Enhance the feature set with contextually meaningful information



Issues Addressed

- Capturing Common Phrases
- Disambiguation
- Improved Feature Set
- Handling Negations

Step 2: Hyper Parameter Tuning

Methods Explored:

1. SMOTE
2. Random Search
3. Bayesian Optimization



Issues Addressed

- Class Imbalances
- Feature Selection
 - Random Search
 - Bayesian Optimization

Step 3: Results

Initial Results:

1. TF-Vectorizer with n-grams helped add contextual understanding
2. Bayesian Optimization yielded the best results



Final Findings

- Bayesian Optimization:
- Training Accuracy: 97.55%
- Test Accuracy: 94.91%

Logistic Regression Classification Report



Test Accuracy: 0.9491470773388199

Test Classification Report:

	precision	recall	f1-score	support
negative	0.76	0.90	0.82	8511
positive	0.98	0.96	0.97	56618
accuracy			0.95	65129
macro avg	0.87	0.93	0.90	65129
weighted avg	0.95	0.95	0.95	65129

BERT with Neural Network - Model 2



Step 1: Pre-Processing & Transformers

TF

- Tokenizer: bert-base-uncased
- Model: TFAutoModel from Hugging Face
- Tokenized reviews with sequence length of 128



Issues Addressed

- Text Data Variability
- Tokenization specific for BERT

Step 2: Hyper Parameter Tuning

Methods Explored:

1. SMOTE
2. Sequential model with multiple layers



Issues Addressed

- Class Imbalances
- Implementation of NN

Step 3: Results

Initial Results:

1. Successfully obtained BERT embeddings for tokenized text
2. Implemented a neural network with dense layers



Final Findings

- Training Accuracy: 91%
- Validation Accuracy: 89%
- Test Accuracy: 88%

BERT with Neural Network Classification Report



	precision	recall	f1-score	support
Negative	0.39	0.65	0.49	8556
Positive	0.94	0.85	0.89	56573
accuracy			0.82	65129
macro avg	0.66	0.75	0.69	65129
weighted avg	0.87	0.82	0.84	65129

CNN with Embeddings- Model 3



Step 1: Pre-Processing & Transformation

Techniques Used:

1. Text Cleaning & Tokenization
2. Padding
3. Exploratory Data Analysis
4. Vocabulary and word count

Issues Addressed

- Text data variability
- Consistent dimensions
- Handling Negations
- Reduced training time

Step 2: Hyper Parameter Tuning

Methods Explored:

1. Addressing class imbalance in model
2. Bayesian optimization
3. Model callbacks
4. Accuracy maximization and loss minimization

Issues Addressed

- Overfitting
- Class imbalance
- Efficient hyperparameter search & training speed

Step 3: Results

Initial Results:

1. Sequential model with multiple layers
2. Model architecture with embeddings and CNN yield high accuracy
3. Tackling class imbalance with class weights

Final Findings

- Training Accuracy: 99%
- Validation Accuracy: 96%
- Test Accuracy: 95%

CNN with Embeddings Classification Report



Test Accuracy: 0.95

Test Classification Report:

	precision	recall	f1-score	support
Negative	0.79	0.88	0.83	8511
Positive	0.98	0.96	0.97	56618
accuracy			0.95	65129
macro avg	0.88	0.92	0.90	65129
weighted avg	0.96	0.95	0.95	65129

LSTM with Attention- Model 4



Step 1: Additional Pre-Processing

Techniques Used:

1. All of the formerly mentioned pre-processing
2. Embedding Lengths
3. Class Weights
4. Undersampling

Issues Addressed

- Optimizing encoding
- Balanced the classes to mitigate bias in training

Step 2: Hyper Parameter Tuning

Methods Explored:

1. Embedding Layers & pre-trained embedding
2. (Multi-head) Attention Layers
3. Bidirectional LSTM layers
4. Pooling

Issues Addressed

- Testing various NN architectures to see what mechanisms help

Step 3: Results

Initial Results:

1. Embedding layers immediately boosted accuracy over dense NN
2. LSTM + attention layers helped improve generalization to unseen data

Final Findings

- Training Accuracy: 96%
- Test Accuracy: 91%

LSTM with Attention Classification Report

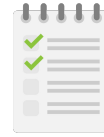


Validation Accuracy: 0.9146652039337069

Validation Classification Report:

	precision	recall	f1-score	support
0.0	0.92	0.91	0.91	12652
1.0	0.91	0.92	0.92	12871
accuracy			0.91	25523
macro avg	0.91	0.91	0.91	25523
weighted avg	0.91	0.91	0.91	25523

NeurIPS 2021 Paper Checklist Guidelines



1. The purpose of this project is to gain an understanding of how to conduct sentiment analysis (POC).
2. Our experimentation phase is well documented
3. Considerable parts of our workflow depend on having access to a GPU.
4. Dataset was open source and backed by a trusted source.
5. We read through the ethics review guidelines and ensured that the project conforms to them.

Conclusion



Key Results

Context is
extremely
important!!!



What We Learned

Task was
computationally
and time
intensive



If we had more time...

Translating lower
occurrence reviews
to foreign languages
and back to
augmented data

